MEDPLUS
PHARMACY POS

Next-Generation Point of Sale & FEFO Inventory Engine
for Healthcare Retail in Kenya

PRESENTER

Ralph Osiko

Reg. No: 23/08138

INSTITUTION

KCA University

School of Technology

SUPERVISOR

Mr. Jacob Oloo

PRESENTATION ROADMAP

Outline

1 Introduction

2 Background

3 Problem Statement

4 System Objectives

5 Significance of Study

6 Literature Review

7 Methodology

8 Results & Achieved Objectives

CHAPTER 1

Introduction

What is MedPlus Pharmacy POS?

MedPlus Pharmacy POS is a web-based Point of Sale system specifically designed for community pharmacies and retail healthcare outlets in Kenya. It integrates sales processing, inventory management, prescription verification, and mobile money payments into a single, unified platform.

Target Market

Community pharmacies and retail healthcare outlets across Kenya serving the rapidly expanding Ksh 120B+ pharmaceutical market.

Technology Stack

Go 1.22+ backend, PostgreSQL 16/18 database, and vanilla HTML5/CSS3/JavaScript frontend for maximum performance.

Market Opportunity

Kenya's community pharmacy sector is growing rapidly, yet most outlets still rely on manual paper-based systems.

Unique Value

Zero recurring license fees, native M-Pesa integration, and automated FEFO expiry control — unmatched in the market.

CHAPTER 2

Background

The Kenyan Pharmaceutical Landscape

Kenya's retail pharmaceutical market is valued at over Ksh 120 billion and continues to grow as community healthcare becomes more accessible. However, the majority of community pharmacies still operate using manual paper ledgers, handwritten receipts, and cash-based transactions.

Ksh 120B+	500ms	11	51
Market Size	Checkout Speed	DB Tables (3NF)	Tests Passed

Why This Matters

The Pharmacy and Poisons Board (PPB) of Kenya mandates strict controls on prescription-only medications and expiry management. Current manual systems make compliance difficult, leading to regulatory risks, patient safety concerns, and financial losses. Existing ERP solutions are either too expensive ($1,500+/year) or lack native M-Pesa integration essential for the Kenyan market.

CHAPTER 3

Problem Statement

Problem 01 — Expired Drug Dispensing Risk

Manual paper ledgers cannot track batch shelf-life across hundreds of medicines, risking accidental dispensing of expired drugs and PPB penalties.

Problem 02 — Stock Shrinkage & Loss

Lack of real-time stock deduction and automated reorder alerts leads to inventory shrinkage, theft, and revenue leakage.

Problem 03 — Rx Non-Compliance

Dispensing Prescription-Only Medication (POM) without verified doctor details risks patient safety and regulatory action.

Problem 04 — Slow Checkout & Cash Errors

Manual cash calculations and manual M-Pesa status checks slow down sales processing and cause daily balancing errors.

Core Problem

There is no affordable, locally-optimized pharmacy POS solution in Kenya that combines automated FEFO expiry control, prescription verification, and native M-Pesa integration while remaining license-free.

CHAPTER 4

System Objectives

Main Objective

To design, implement, and evaluate a web-based Pharmacy POS System for MedPlus Pharmacy that automates sales checkout, enforces FEFO inventory expiry control, validates doctor prescriptions, and processes M-Pesa payments.

Specific Objectives

Dynamic Auth & RBAC

Bcrypt password protection with role-based access for Admin, Pharmacist & Cashier

FEFO Inventory Control

Batch tracking & automated reorder alerts for expiry management

Fast POS Checkout

Barcode scanning & M-Pesa STK Push API integration

Prescription Verification

Validate doctor & patient details prior to sale of POM drugs

Supply Chain Mgmt

Manage supplier profiles & Purchase Order receiving

Empirical Validation

Achieve 100% Pass across GUI, DB & Output benchmarks

CHAPTER 5

Significance of the Study

This project addresses a critical gap in Kenya's healthcare technology infrastructure. The significance extends across multiple stakeholders:

For Pharmacy Owners

Eliminates stock shrinkage and expired drug losses. Reduces checkout time to under 30 seconds per customer. Zero recurring license fees replace expensive $1,500+/year ERP subscriptions.

For Pharmacists

Automated FEFO controls ensure oldest batches are dispensed first. Prescription verification enforces PPB compliance with minimal manual effort.

For Patients

Improved patient safety through verified prescription dispensing. Faster checkout means shorter queues. Assurance that purchased medications are within valid expiry dates.

For the Industry

Provides a replicable, affordable model for digitizing community pharmacies across East Africa. Demonstrates that locally-built, license-free solutions can compete with expensive international ERPs.

CHAPTER 6

Literature Review

A review of existing pharmacy management systems reveals significant gaps that MedPlus POS addresses:

OpenMRS / Bahmni

Java Enterprise-based hospital management system. Heavy architecture, high setup complexity. No native M-Pesa integration. Designed for hospitals, not retail pharmacies.

Not Suitable

Odoo Pharmacy Module

Python/PostgreSQL ERP with pharmacy extension. FEFO available only in Enterprise Edition. M-Pesa requires paid third-party plugin. High monthly per-user fees.

Partial Fit

QuickBooks POS

Desktop C#/SQL solution for general retail. No pharmacy-specific features. FIFO/LIFO only — no FEFO expiry control. No M-Pesa integration.

Not Suitable

MedPlus POS (This Project)

Go Micro-Monolith with native M-Pesa Daraja 2.0 API, automated FEFO & alerts, prescription verification, and zero recurring license fees.

Optimal Fit

Research Gap

No existing solution combines all four critical requirements for Kenyan pharmacies: native M-Pesa integration, automated FEFO expiry control, prescription verification, and zero recurring costs.

CHAPTER 7

Methodology

This project followed the Agile Software Development Life Cycle (SDLC) with iterative prototyping, combined with empirical validation through structured testing.

Requirements Analysis

Stakeholder interviews with community pharmacists. Reviewed PPB regulations, M-Pesa Daraja 2.0 API documentation, and FEFO inventory best practices.

System Design

3-tier micro-monolith architecture. 11 normalized PostgreSQL tables in 3NF. Entity-Relationship diagrams and API endpoint specifications.

Implementation

Go 1.22+ backend with net/http router. Vanilla HTML5/CSS3/JS frontend. Bcrypt authentication, M-Pesa STK Push handler, FEFO rules engine.

Testing & Validation

51 comprehensive test cases: 24 GUI tests, 15 DB tests, 12 Output tests. All achieved 100% PASS rate across validation tiers.

Development Tools & Technologies

Backend	Database	Frontend	Payment
Go 1.22+	PostgreSQL 16/18	HTML5/CSS3/JS	M-Pesa Daraja 2.0

SYSTEM DESIGN

High-Concurrency 3-Tier Architecture

Client Tier Single-Page App SPA	Vanilla HTML5/CSS3/JavaScript with Inter & Outfit fonts. Zero heavy client frameworks for lightning-fast DOM rendering and minimal bundle size.
↓	
Application Tier Go RESTful Micro-Monolith Go 1.22+	Compiled binary backend using net/http router, bcrypt auth, M-Pesa STK Push handler, FEFO rules engine. Response time < 500ms.
↓	
Database Tier PostgreSQL 16/18 Relational DB	11 normalized tables in 3NF — users, products, sales, prescriptions, suppliers, mpesa_payments, and more. ACID transactions.

PRODUCT FEATURES

End-to-End Pharmacy Operational Modules

Authentication & RBAC

Bcrypt password protection & dynamic permission matrix across Admin, Pharmacist, and Cashier roles.

Prescription Dispensing

Doctor Rx validation workflow preventing unverified dispensing of prescription-only medications.

FEFO Inventory Control

Real-time stock deduction, batch expiry tracking, and automated reorder warnings.

Supply Chain & POs

Supplier management and Purchase Order receiving into inventory with full traceability.

POS Checkout & M-Pesa

Barcode cart management, cash change calculation, and instant M-Pesa Daraja 2.0 STK Push.

Security Audit & DB Token

Timestamped security logs and 10-minute access token for secure DB access.

MARKET ANALYSIS

MedPlus POS vs Existing Solutions

Feature / Capability	OpenMRS / Bahmni	Odoo Pharmacy	QuickBooks POS	MedPlus POS
Architecture & Stack	Java Enterprise (Heavy)	Python / PostgreSQL	Desktop C# / SQL	**Go Micro-Monolith (Fast)**
M-Pesa STK Push API	No (Custom plugin needed)	Paid Third-Party Plugin	No Native Integration	**Native Daraja 2.0 API**
FEFO Expiry Control	Partial Batch Tracking	Yes (Enterprise Edition)	No (FIFO/LIFO only)	**Automated FEFO & Alerts**
Recurring Licensing Fee	Open Source (High Setup)	High Monthly/User Fee	Perpetual / Legacy Fee	**Zero Recurring License Fee**

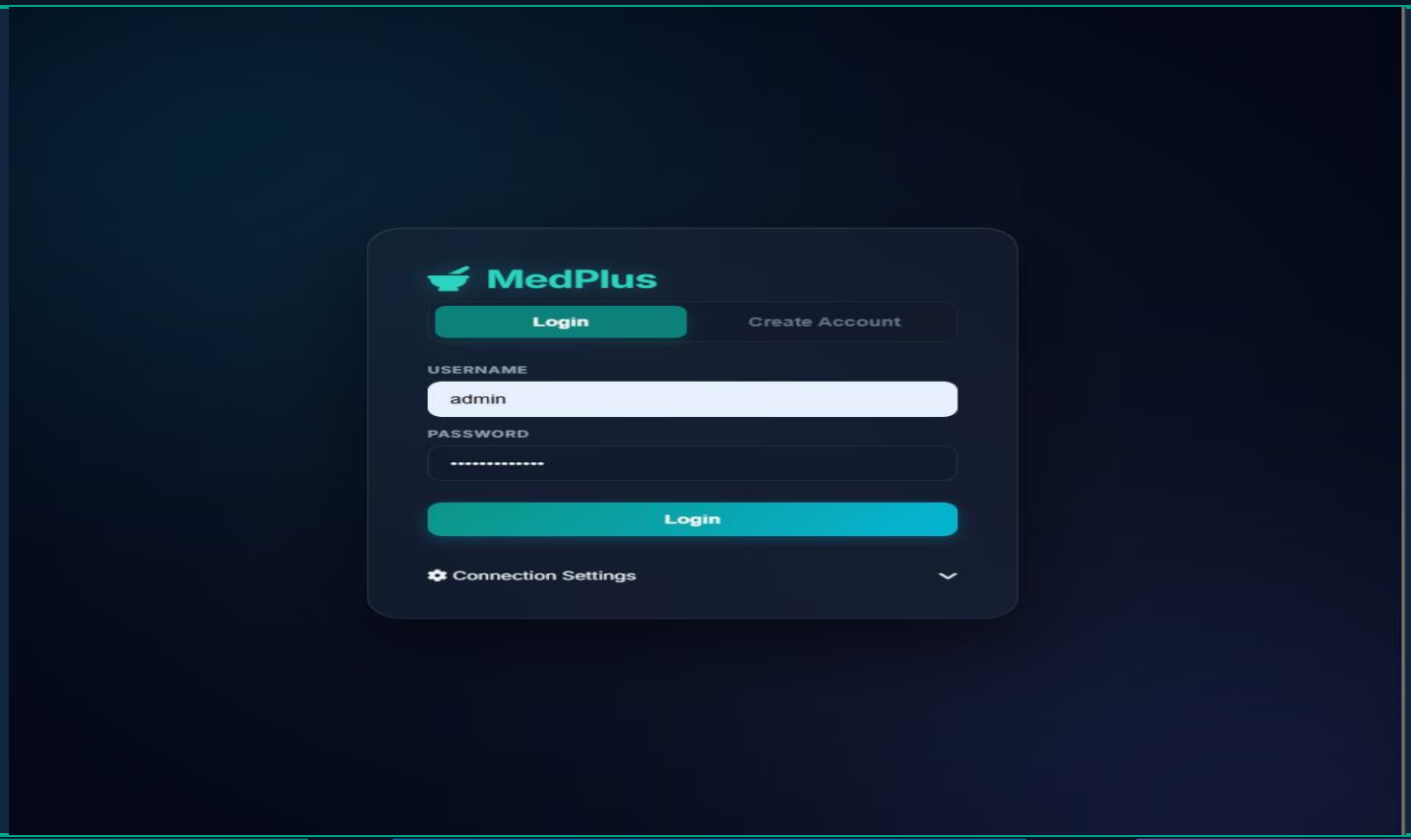
Key Difference

MedPlus POS combines Kenyan payment needs, pharmacy compliance, and low ownership cost in one tailored platform.

System Screenshots — Login & Authentication

Login & Authentication Module

Secure login with Bcrypt password hashing and role-based access control



Feature

Bcrypt password hashing

Roles

Admin, Pharmacist

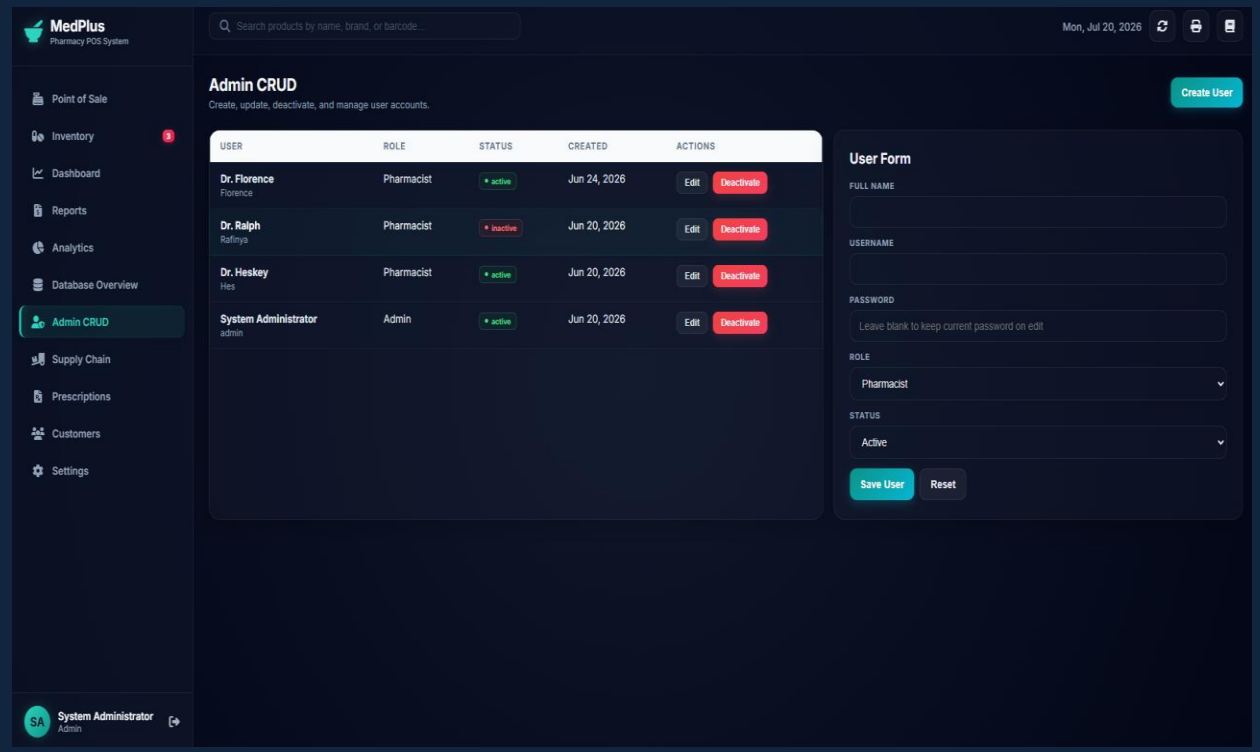
Security

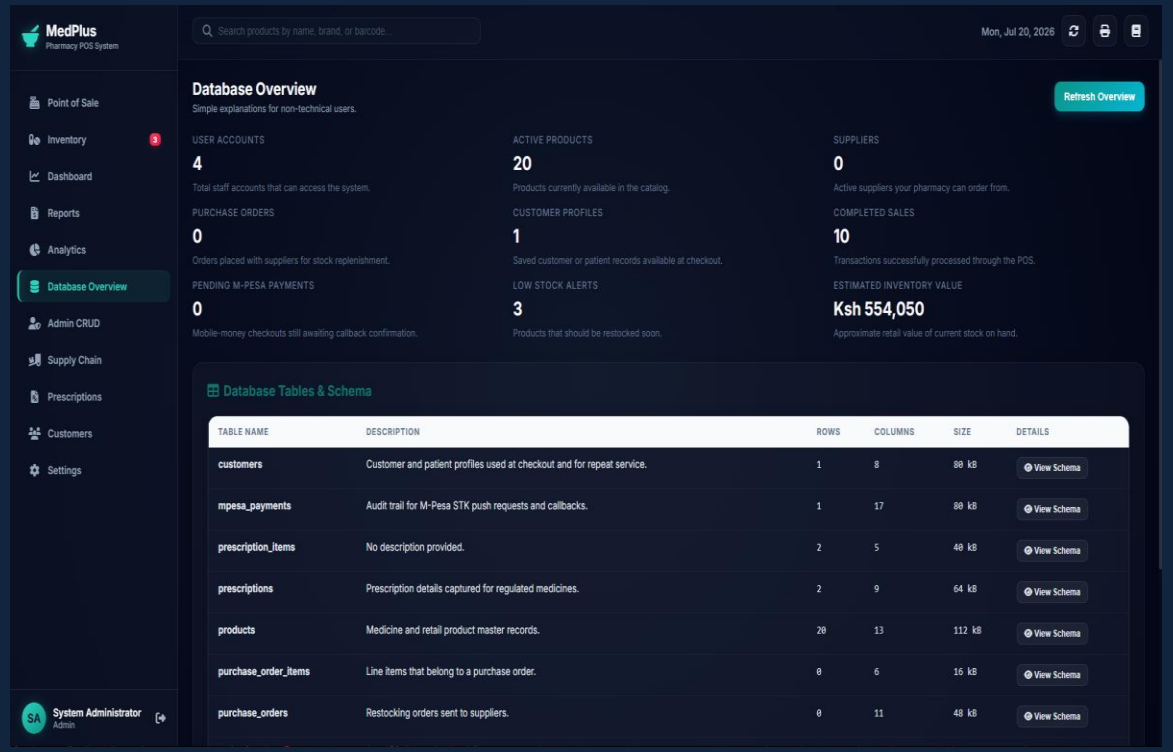
10-min access token expiry

System Screenshots — Dashboard & Analytics

Admin Dashboard Module

Real-time sales analytics, inventory overview, and user activity monitoring





Charts

Sales trends & revenue

Alerts

Low stock & expiry warnings

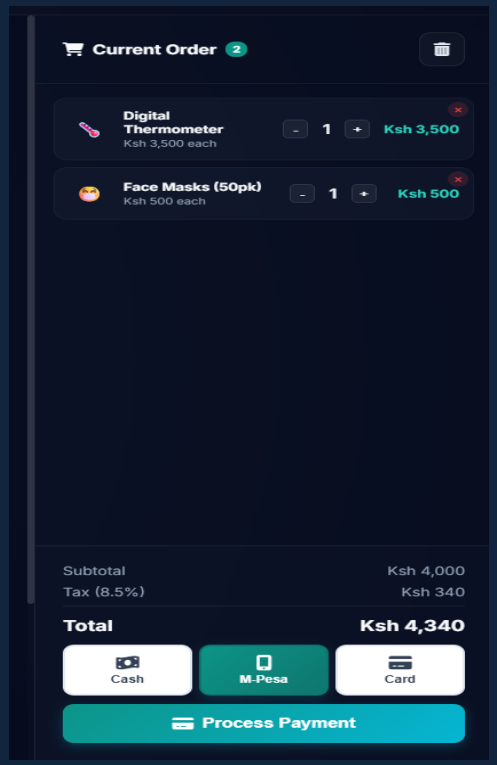
Audit

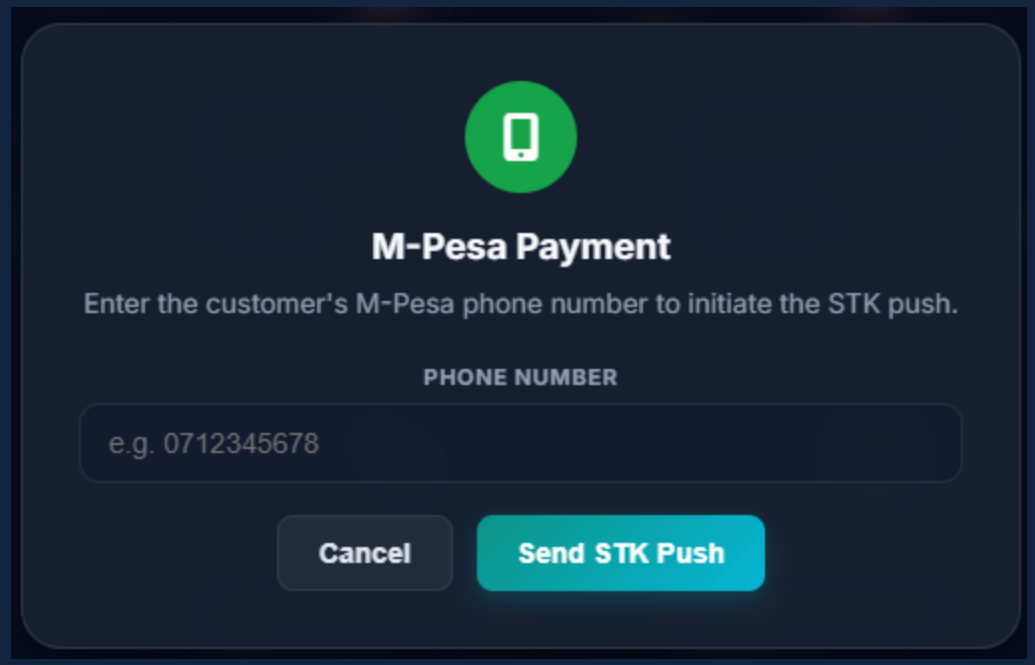
Timestamped activity logs

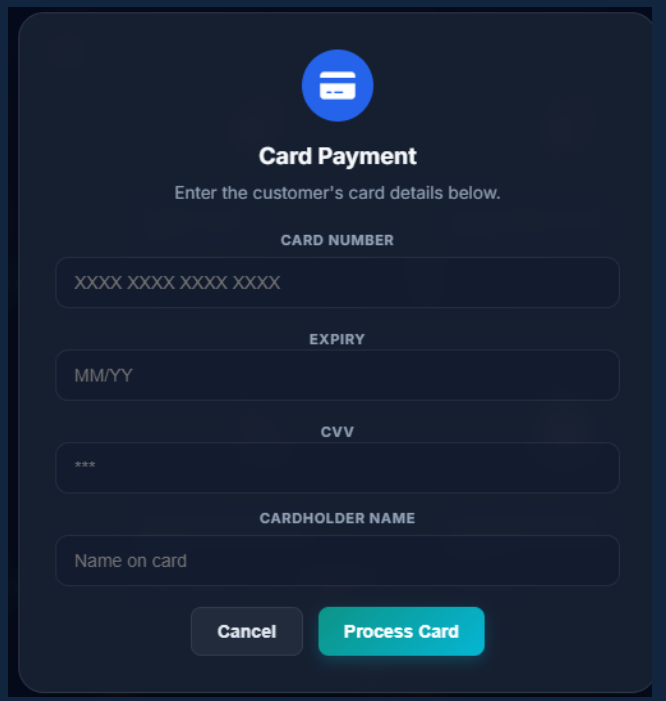
System Screenshots — POS Checkout

Point of Sale Checkout Module

Cart management, and M-Pesa STK Push payment processing







Speed

< 30 seconds per customer

Payment

M-Pesa STK Push + Cash

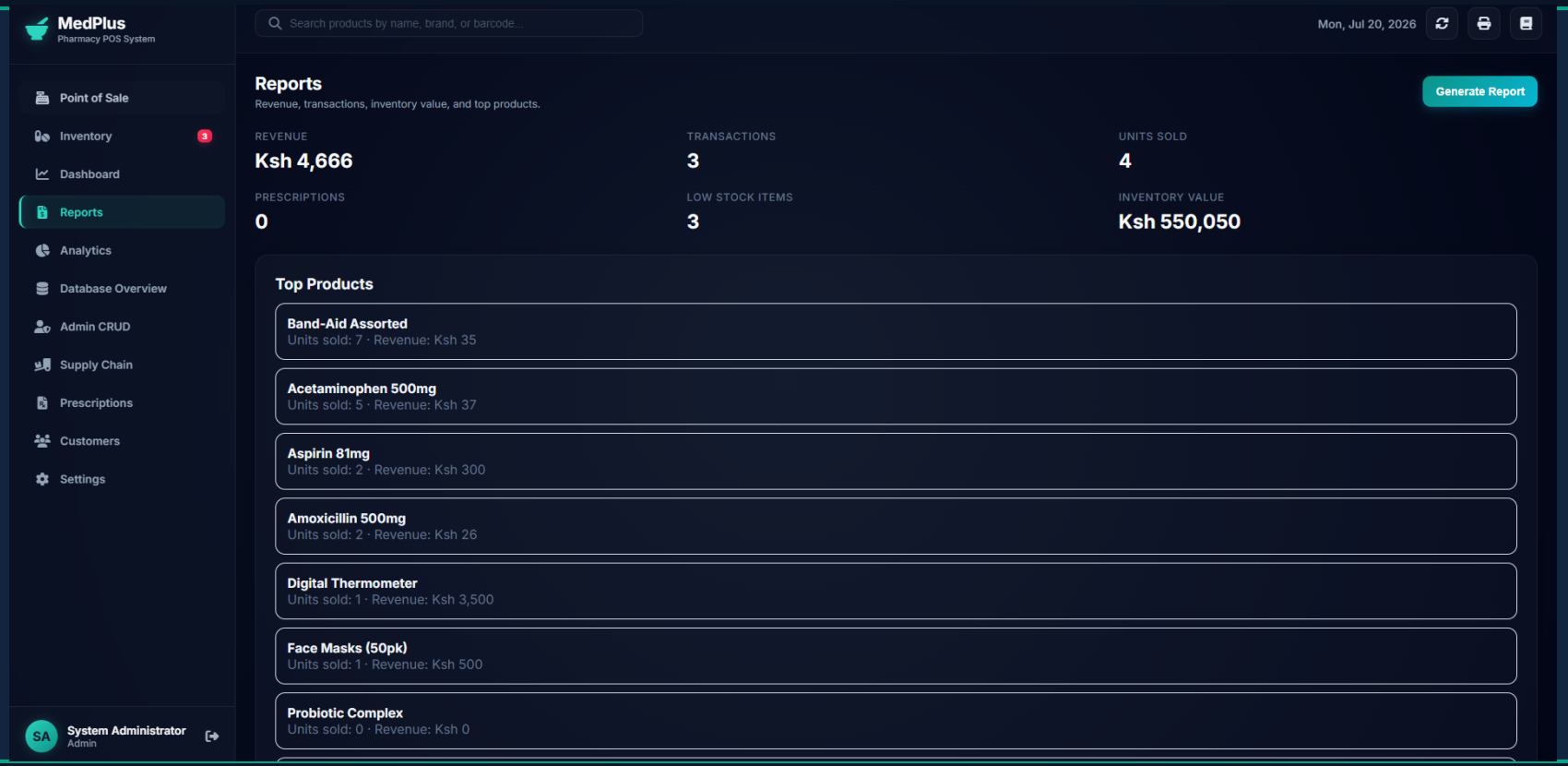
Receipt

16% VAT auto-calculated

System Screenshots — FEFO Inventory

FEFO Inventory Control Module

Batch tracking, expiry monitoring, and automated reorder alerts



FEFO

First-Expired-First-Out

Batches

Unique barcode tracking

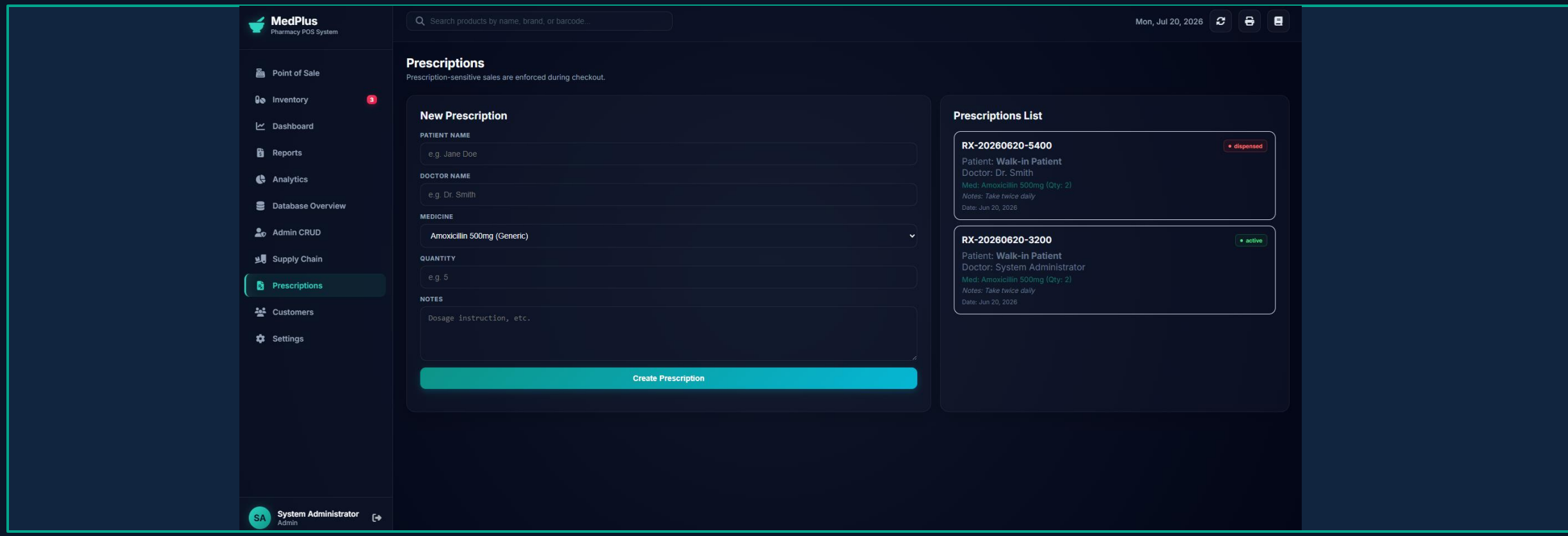
Alerts

Auto reorder warnings

System Screenshots — Prescription Verification

Prescription Dispensing Module

Doctor and patient validation before dispensing Prescription-Only Medications (POM)



Compliance

PPB regulation enforced

Validation

Doctor & patient details

Safety

Prevents unauthorized POM

QUALITY ASSURANCE

100% Pass Rate Across All Validation Tiers

24
Test Cases

GUI Tests

Login redirects, cart updates, Rx modal prompts, M-Pesa spinner, product catalog saves, RBAC

100% PASS

15
Test Cases

DB Tests

UNIQUE barcode constraints, foreign key cascades, atomic checkout transactions, DB security token

100% PASS

12
Test Cases

Output Tests

16% VAT receipt calculations, M-Pesa STK callbacks, daily sales summaries, timestamped audit logs

100% PASS

Total Test Coverage

51 Tests — 100% PASS

Achieved Objectives

All six specific objectives were successfully achieved and validated through empirical testing:

✓ Dynamic Auth & RBAC

Implemented Bcrypt password hashing with Admin, Pharmacist, and Cashier role-based access control. Validated through 24 GUI test cases.

✓ FEFO Inventory Control

Built batch tracking with automated reorder alerts. First-Expired-First-Out logic prevents sale of expired medications. 15 DB tests passed.

✓ Fast POS Checkout

Barcode scanning cart management with M-Pesa STK Push API integration. Checkout speed under 30 seconds per customer.

✓ Prescription Verification

Doctor Rx validation workflow prevents unverified dispensing of POM drugs. Enforces PPB regulatory compliance at point of sale.

✓ Supply Chain Management

Supplier profiles and Purchase Order receiving into inventory with full traceability. 11 normalized PostgreSQL tables in 3NF.

✓ Empirical Validation

51 comprehensive test cases across GUI, Database, and Output tiers. All achieved 100% PASS rate — zero failures.

CLOSING

Summary & Future Roadmap

Project Summary

✓ Successfully developed and tested a Go/PostgreSQL Pharmacy POS system for MedPlus Pharmacy

✓ Automated checkout, prescription validation, and M-Pesa STK Push payment processing

✓ Eliminated expired medicine sales through strict FEFO batch rotation controls

✓ Achieved 100% empirical pass rate across 51 comprehensive test cases

Future Roadmap

AI-Driven Demand Forecasting

Predictive models to forecast seasonal antibiotic and medication demand

Kenya SHA / NHIF Claims API

Direct digital insurance claim preauthorization at point of sale

Multi-Branch Cloud PWA Sync

Offline-first PWA connecting remote pharmacy branches

Thank You — Questions & Discussion

Ralph Osiko | School of Technology, KCA University